

LeetCode Helper - AI-Powered Productivity Enhancement

Overview

LeetCode Helper is a Chrome extension that monitors your interactions within the LeetCode platform and uses machine learning to predict your next actions. By offering timely suggestions via a minimally invasive "Press Tab" UI, the extension helps reduce friction and saves you valuable time when solving coding problems.

Features

1. Tab Suggestions

- Predicts your next action based on interaction patterns
- Shows a subtle overlay with "Press Tab to [action]" when confidence is high
- Automatically executes the action when Tab is pressed

2. Command Bar Autocomplete

- Monitors what you type in search/command bars
- Suggests completions based on your past searches and common patterns
- Uses LLM to generate context-aware suggestions even for new queries

3. Automation Detection

- Identifies repetitive sequences of actions you perform
- Offers to automate these sequences for future use
- Creates shortcuts for multi-step processes you commonly use

4. Contest Reminders

- Learns your contest participation patterns
- Reminds you of upcoming contests at your preferred times
- Gets smarter over time by analyzing when you actually participate

Technical Architecture

The solution consists of three main components:

1. **Chrome Extension:**

- Content Script: Injects UI overlay and monitors user interactions
- Background Script: Processes data and communicates with backend
- Popup UI: Provides user-configurable settings and stats

2. **Python Backend:**

- Flask API: Processes requests from the extension

- Machine Learning: RandomForest classifier for action prediction
- LangChain + OpenAI: Advanced LLM-based predictions and autocomplete
- User Data Store: Persistent storage of interaction patterns

3. **Prediction System:**

- Hybrid approach combining traditional ML with LLM capabilities
- Confidence-based suggestion surfacing
- Continuous learning from user feedback

Implementation Details

Data Collection

The extension monitors:

- Network requests (API calls)
- UI interactions (clicks, form submissions)
- Page navigation events
- Timing patterns (when certain actions are performed)

ML Prediction Pipeline

1. **Feature Extraction:**

- Recent sequence of actions
- Current page context
- Focused element information
- Historical patterns

2. **Prediction Models:**

- Primary: RandomForest classifier trained on user data
- Backup: LLM-based prediction for novel situations
- Confidence thresholds to ensure high-quality suggestions

3. **Action Execution:**

- Mapping predictions to executable actions
- Handling navigation, form submissions, and API calls

Extensibility

The architecture is designed to be extensible:

- New prediction features can be added without changing the core system
- Different SaaS applications could be supported with minimal changes
- The ML models can be swapped or enhanced as needed

Future Improvements

1. **Enhanced UI Integration:**

- More native-feeling suggestions integrated into LeetCode's UI
- Keyboard shortcut customization

2. **Advanced ML Capabilities:**

- Incorporate transformer-based models for sequence prediction
- Add reinforcement learning from user feedback
- Implement collaborative filtering using anonymized pattern data

3. **Expanded Feature Set:**

- Code template suggestions based on problem type
- Time management recommendations
- Integration with other development tools

4. **Scalability:**

- Cloud-based backend for multi-user support
- Efficient data storage for years of interaction history
- Privacy-preserving federated learning

Installation and Setup

1. Clone the repository
2. Set up the Python backend:

```
pip install -r requirements.txt
export OPENAI_API_KEY="your_api_key_here"
python app.py
```

3. Load the Chrome extension:

- Open Chrome and navigate to <chrome://extensions/>
- Enable Developer Mode
- Click "Load unpacked" and select the extension directory

Usage

1. Navigate to LeetCode and use the platform as normal
2. Watch for the subtle "Press Tab" suggestions
3. When you see a suggestion that matches your intent, press Tab
4. The extension will automatically execute the action for you
5. Access settings via the extension icon in Chrome's toolbar

Privacy Considerations

- All data is stored locally by default
- OpenAI API calls are made without sharing personal identifiers
- You can clear collected data at any time from the extension popup

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Developed as a demonstration of predictive UI for enhancing SaaS product experiences.